ISSN 0038-0946, Solar System Research, 2007, Vol. 41, No. 1, pp. 28–41. © Pleiades Publishing, Inc., 2007.
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Atmospheric Erosion and Replenishment Induced by Impacts of Cosmic Bodies upon the Earth and Mars

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Received November 11, 2005

Abstract—The atmospheric erosion induced by impacts of cosmic bodies with sizes from ~100 m to 10 km is calculated for the Earth with its present atmosphere and for Mars with a dense carbon dioxide atmosphere that could be at the early stages of planetary evolution. Numerical results are compared to simple analytic models and calculations performed by other authors; approximate formulas are suggested. The evolutions of early atmospheres, which could exist at the late stage of the planetary accumulation, are numerically simulated using an integral model of impact-induced atmospheric erosion and replenishment in the approximation of a one-component atmosphere with a composition determined by the basic atmophile component of the bodies falling onto the planet.

PACS numbers: 96.12.je; 96.12.ke

DOI: 10.1134/S0038094607010030

INTRODUCTION

A loss of the atmospheric gas portions that gain the velocity above the escape velocity and fly away into space can lead to the erosion (depletion) of the atmosphere. Such losses may accompany asteroid and comet impacts upon planets (Kaula, 1968; Cameron, 1983; Hunten et al., 1989; Ahrens, 1993). But it is still not clear how efficient the impact mechanism of the atmospheric erosion is. If a large body with a size of 10 km or more and a high velocity strikes the Earth in its present state, the energy and momentum are transferred to the atmospheric gas mainly during the expansion of the jet of vapors from the impactor and planet materials and during the jet interaction with the atmosphere (Hunten et al., 1989; Vickery and Melosh, 1990). Some simple models were suggested to allow for this interaction (Kaula, 1968; Melosh and Vickery, 1989; Vickery and Melosh, 1990). They were applied to study the evolution of the Martian atmosphere (Melosh and Vickery, 1989; Zahnle, 1993). In the simplest model (Kaula, 1968; Melosh and Vickery, 1989), the entire atmospheric mass above the plane tangent to the planet surface is taken away if the mass and the velocity of the impactor are above certain thresholds; no atmospheric erosion occurs for smaller masses and velocities of impacting bodies (a tangent-plane model). In a more complicated semianalytic "sector" model (Vickery and Melosh, 1990), the interaction between the vapor jet and the atmosphere is taken into account approximately in the intermediate size range. However, on the one hand, these models are too schematic and do not provide an acceptable accuracy (as demonstrated, e.g., by Vickery (1994) and Svetsov (2000)). On the other hand, impacts are accompanied not only by the escape of a portion of the atmospheric gas, but also by a replenishment of the atmosphere by atmophile components (carbon and its compounds, nitrogen, oxygen, noble gases, hydrogen) contained in asteroids and comets (Zahnle et al., 1992). That is why the ratio of the lost mass of the atmospheric gas to the impactor mass is so important. This ratio can grow with decreasing size of a body. However, the above models are not appropriate for relatively small bodies, when the wake formed behind the falling body in the atmosphere plays a significant role.

Numerical simulations of vertical impacts of 20-km-diameter bodies (Svetsov, 2000) demonstrated that the masses lost by an atmosphere identical to the present-day Earth's atmosphere are much smaller than those predicted by the "sector" model (Vickery and Melosh, 1990) and make only fractions of a percent of the asteroid or comet mass; thus, they are less than the mass of volatiles contained in carbonaceous chondrites and comets and joining the atmosphere after the impact. For a 2-km body, the atmospheric losses were estimated at about 10% of the impactor mass. But these losses were overestimated because of the coarse difference grid applied in computations. This follows from comparing the above results with more accurate simulations of impacts of bodies with diameters of 1, 3, 10, and 30 km performed by Shuvalov and Artemieva (2000, 2002). According to their computations, the loss of the atmospheric mass induced by impacts of comets from 1 to 3 km in diameter at a velocity of 50 km/s is only about 1% of the comet mass. The loss is about 0.1% of the impactor mass after impacts of asteroids of the same size with a velocity of 20 km/s.

In the case of vertical impacts of bodies smaller than 1 km in size, the atmospheric mass is lost because the

atmosphere is heated by the falling cosmic body and the air is accelerated upward along the tenuous wake left by the body. The analytic model suggested for this case (Svetsov, 2000) shows that the atmospheric gas can be ejected more efficiently due to impacts of bodies with diameters of hundreds of meters. However, this model needs further verification. No numerical simulations of the atmospheric erosion were performed for impacts of such small bodies with the exception of the study of Nemchinov and Shuvalov (2003), who estimated the losses of the Earth's atmosphere induced by vertical impacts of 200- and 300-m comets moving at a velocity of 50 km/s to be 6 and 8% of the comet mass, respectively. These values are much higher than the relative losses obtained for the impact of a 1-km comet by Shuvalov and Artemieva.

In the present study, we performed numerical simulations of impacts of comets and asteroids with a size of 10 km or less, down to diameters of about 100 m, when the lost atmospheric mass becomes small. The final aim of such computations is to construct models of the atmospheres' evolution. At present, the planetary atmospheres are not subjected to noticeable impact erosion and replenishment, because the flux of cosmic bodies has been relatively small during the last 3.5 billion years. But at the latest stage of planetary accumulation, when the flux of falling bodies was rather large, parameters of early atmospheres were presumably determined mainly by impact degassing (Florenskii, 1965; Gerasimov and Mukhin, 1979) and erosion. The modern atmospheres of the Earth and Mars differ from the early ones, which were possibly denser, since, according to the estimates of Walker (1985), the partial carbon dioxide pressure of the early Earth's atmosphere was about 10 bar. Various other processes of replenishment (magma differentiation and volcanism) and sink could have affected the evolution of atmospheres after the accretion had ceased; in particular, carbon dioxide could have gone with time to carbonates or gas hydrates.

In the present study, the evolution of the early atmospheres of the Earth and Mars at the late stage of the planets' accumulation is calculated using a simple model of the one-component atmosphere. It is assumed that the primordial hydrogen-containing atmosphere, arising from the protoplanetary cloud, had been lost, and the atmospheric composition was determined by the basic atmosphere-forming component of colliding bodies, carbon in this case.

MODELING OF IMPACTS AND ATMOSPHERIC LOSSES FOR THE PRESENT-DAY EARTH

Vertical impacts were numerically simulated by solving two-dimensional gas dynamic equations in cylindrical coordinates via the numerical SOVA method (Shuvalov, 1999; Shuvalov et al., 1999), in which different materials were characterized by their own concentrations and the boundaries between them were approximately taken into account, when these materials flew from one computational cell to another. The strength was not taken into account in the equations, because the goal of our computations was not to model crater formation but to find the mass of the lost material, which is determined by evaporation at the initial stage of the impact, when the pressure behind the shock wave propagating along the target is high. Simulations of impacts of comets and stony bodies and calculation of the lost atmospheric mass were performed as in the papers of Shuvalov and Artemieva (2000, 2002). The only difference was in the equation of state for water, which was computed for modeling of comet impacts and presented in table form by Smetannikov (Teterev et al., 1993). The equation of state for granite in Tillotson's form (Tillotson, 1962; Melosh, 1989) was used for stony bodies. For air, we applied the equation of state in table form (Kuznetsov, 1965) as in the above-mentioned studies of Shuvalov and Artemieva. The difference grid has the same resolution of 20 grid points across the body radius, and the cells became larger to the periphery of the computation domain. The impacting body was assumed to be cylindrical in shape with a height equal to its diameter. (The body shape is not of principal importance, but computations are less complicated for a cylindrical body and it is easier to compare the results with an analytic model of the motion of a fragmented body in the atmosphere.)

The atmosphere was assumed to be exponential with the scale height H = 7.5 km and the surface pressure P =1 bar. A body was considered to enter the atmosphere at a height of 150 km, to move downward, and, being distorted to some extent by aerodynamic forces, to hit the oceanic surface in the case of a falling comet (with a density of 1 g/cm^3) or a stony target (with a near-surface density of 2.7 g/cm^3, which is similar to the asteroid density). The thickness of the target was chosen large enough so that the lower boundary did not affect the output parameters of the problem. We modeled the impacts of comets and asteroids, which entered the atmosphere at velocities from 20 to 50 km/s and from 15 to 20 km/s, respectively.

Figure 1 illustrates an impact of an icy body with a diameter of 100 m and an entry velocity of 50 km/s. While falling in the atmosphere, the body gradually disintegrates and the cross-section diameter of the disintegrated mass or a swarm of fragments increases. The fragments occupy a domain about 400 m in diameter at a height of 14 km (Fig. 1a); the shock wave is a little ahead of the leading fragments in this case. The cometary material is spread over a region 2 km in diameter and about 2 km in height at the moment when the shock wave reaches the surface (Fig. 1b). About 80% of the initial kinetic energy is spent on the air and vapor heating before the impact on the surface. The crater formed in the water is small, but a column of water rises at the impact site as high as 20 km (Figs. 1d and 1e). The heated air is accelerated and ejected along the wake. The ejected air reaches a height of about 10 km

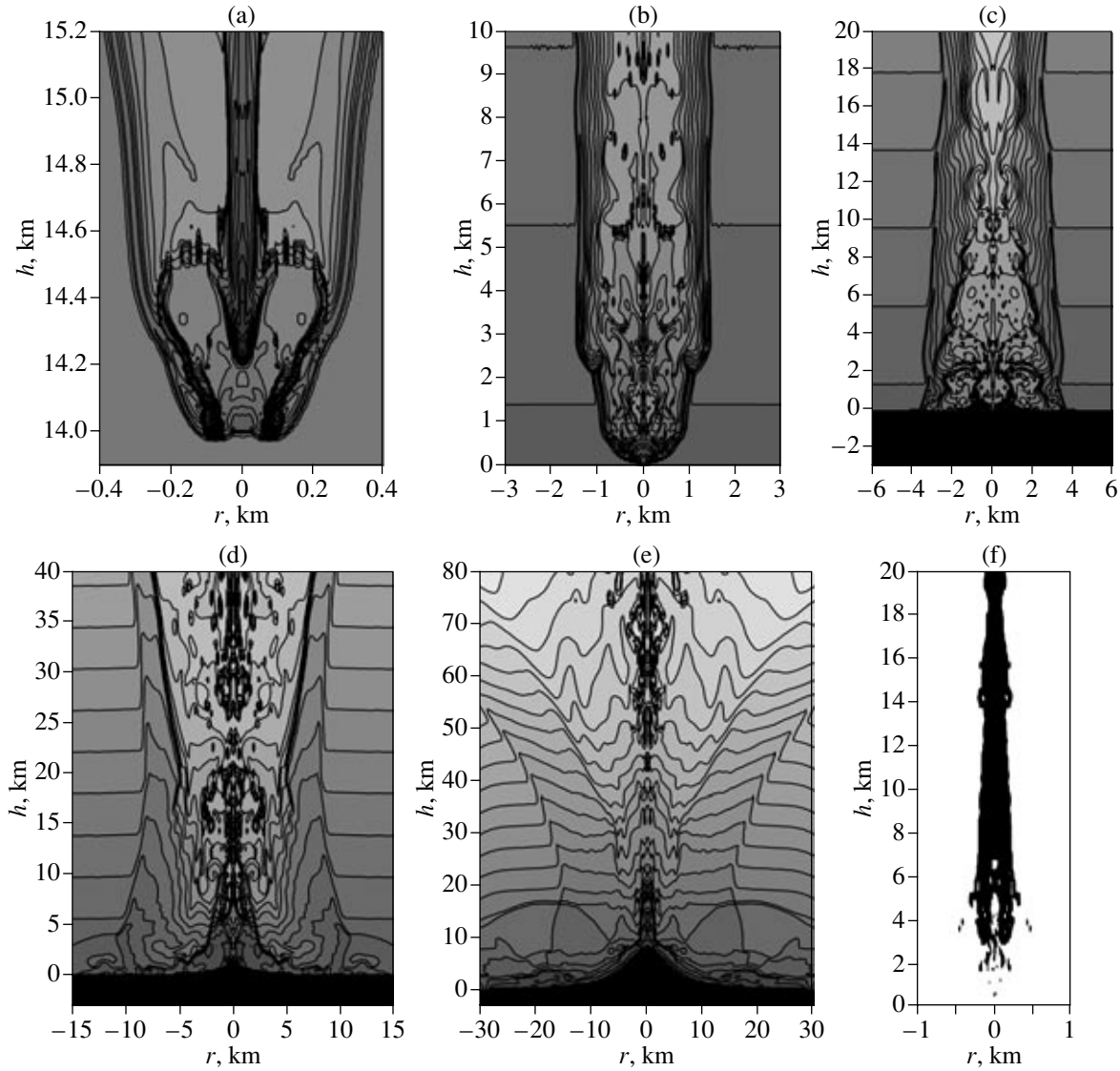

Fig. 1. Isolines of the common logarithm of density at a vertical impact of an icy body with a diameter of 100 m and a velocity of 50 km/s for several time moments: (a) when the body, which had been disintegrated in its flight, arrived at a height of 14 km; (b) when fragments of the body reached the Earth's surface; (c) 1 s after the impact upon the surface; (d) 10 s after the impact; (e) 30 s after the impact. The pre-impact position of the air mass that leaves the planet after the impact is shown in dark in (f). The height h is measured from the Earth's surface and r is the distance from the axis of symmetry.

1 s after the impact on the water (Fig. 1c) and heights from tens to hundreds of kilometers 10 s after the impact (Fig. 1d). Thirty seconds after the impact, the low-velocity part of the ejecta from the crater is below 100 km; it falls back onto the Earth later. Acoustic-gravity waves are propagated in the atmosphere after the impact.

Table 1 presents characteristic parameters for impacts of different-sized comets at a velocity of 50 km/s. The air mass M_{ae}, which gains a velocity of more than 11.2 km/s, grows with increasing impactor mass M, but the ratio M_{ae}/M decreases with increasing body diameter D, inversely proportional to D for large impactors, as the lost air mass is proportional to HD^2 in the limit and the body mass M is proportional to D^3. At small D, the ejection efficiency M_{ae}/M additionally grows due to increasing area of the cross-section of the fragmented body, increasing wake diameter, and growing area of the cross-section of the heated air column. But M_{ae}/M has a maximum at D = 100 m, because for smaller sizes the wake becomes narrow as compared to H and long as compared to the region of energy release, thus preventing the ejection. Before the impact, the air mass that is later lost occupies some volume (ejection region), which is approximately cylindrical at small D (Fig. 1f) and has the shape of a frustum cone at large D. The height of the center of gravity H_e, by which the ejected air mass is characterized before the impact, is

Table 1. Characteristics of impacts of different-sized comets entering the atmosphere at a velocit $V = 50$ km/s

D, km	M, g	D_{max}/D	E_s/E	M_{ae}/M	M_{ao}/M	M_{pe}/M	M_{te}/M	M_{po}/M	M_{to}/M	H_e/H
0.075	3.3×10^{11}	3.5	0.13	0.18	1.8	0.0018	0	0.04	0	2.8
0.1	7.8×10^{11}	3.5	0.21	0.77	1.4	0.015	0	0.081	0	1.1
0.15	2.6×10^{12}	3.5	0.68	0.51	0.49	0.19	0.15	0.13	0.065	0.8
0.2	6.3×10^{12}	3.2	0.83	0.36	0.27	0.24	0.18	0.24	0.34	0.92
0.3	2.1×10^{13}	1.6	0.92	0.21	0.14	0.27	0.16	0.31	0.57	1.1
0.5	9.8×10^{13}	1.5	0.96	0.091	0.06	0.29	0.1	0.37	0.91	1.1
1	7.8×10^{14}	1.3	0.985	0.018	0.013	0.36	0.18	0.34	0.67	1.5
3	2.1×10^{16}	1	0.997	0.0056	0.011	0.25	0.25	0.59	0.99	3.9
10	7.8×10^{17}	1	1	0.0027	0.0067	0.27	0.44	0.56	1	3.2

Note: The following notation is used: D is the body diameter; M is the body mass; D_{max} is the maximal diameter of the fragmented body during its fall in the atmosphere; $E = MV^2/2$ is the kinetic energy of the body when it enters the atmosphere; E_s is the kinetic energy of the body, the products of its ablation, and the air when the body or its remainders reaches the planetary surface; M_{ae} is the lost atmospheric mass (getting the velocity above the escape one); M_{ao} is the atmospheric mass ejected to high altitudes with velocities between the first cosmic velocity and the escape velocity; M_{pe} is the lost mass of the body; M_{te} is the lost mass of the target; M_{po} is the body mass, which gains the velocity between the first cosmic and escape velocities; M_{t0} is the target mass getting the velocity between the first cosmic and escape velocities; H_e is the average height, at which the lost part of the atmosphere was placed before the impact.

Table 2. Characteristics of impacts of comets entering the atmosphere at a velocity of 70 km/s

D, km	M, g	D_{max}/D	E_s/E	M_{ae}/M	M_{ao}/M	M_{pe}/M	M_{te}/M	M_{po}/M	M_{to}/M	H_e/H
0.15	2.6×10^{12}	3.3	0.68	1.2	1.1	0.52	0.79	0.39	1.1	0.92
0.3	2.1×10^{13}	1.6	0.93	0.29	0.19	0.6	0.96	0.39	1.5	1.1
0.5	9.8×10^{13}	1.5	0.95	0.12	0.063	0.7	1	0.23	1.7	1.2
1	7.8×10^{14}	1.3	0.985	0.028	0.093	0.68	1	0.057	2.3	1.4
3	2.1×10^{16}	1	0.997	0.019	0.017	0.84	1.6	0.012	2.2	4.3
10	7.8×10^{17}	1	1	0.0086	0.0039	0.87	1.6	0.0036	1.8	2.3

about H at $D < 1$ km and grows to several H with increasing D as the shape of the ejection region changes.

The characteristic parameters for impacts of comets moving at other velocities are shown in Tables 2–4. In addition to the lost atmospheric mass, the tables also present the calculated losses of impactor and target masses as well as the masses that gain velocities of more than 7.9 km/s and are potentially able to enter orbits around the Earth. While falling in the atmosphere, a meteoroid is squashed by aerodynamic loadings and, as seen from Tables 1–4, its diameter can increase by a factor of 3 to 4. The efficiency of the atmospheric ejection decreases with decreasing impact velocity and becomes very low, about 10^{-3}, at $V = 20$ km/s. A considerable mass of the impactor and target materials is lost at higher velocities of 50 and 70 km/s. However, even at $V = 30$ km/s, these losses become negligi-

Table 3. Characteristics of impacts of comets entering the atmosphere at a velocity of 30 km/s

D, km	M, g	D_{max}/D	E_s/E	M_{ae}/M	M_{ao}/M	M_{pe}/M	M_{te}/M	M_{po}/M	M_{to}/M	H_e/H
0.15	2.6×10^{12}	4	0.67	0.046	0.096	0.0015	0	0.023	0	1.3
0.2	6.3×10^{12}	2.8	0.84	0.063	0.16	0.018	0	0.074	0.072	1.4
0.3	2.1×10^{13}	1.7	0.93	0.051	0.095	0.028	0.0025	0.071	0.052	1.2
0.5	9.8×10^{13}	1.6	0.96	0.026	0.027	0.036	0.0009	0.065	0.024	1.1
1	7.8×10^{14}	1.3	0.985	0.012	0.0061	0.031	0	0.132	0.016	1.6
3	2.1×10^{16}	1.05	0.997	0.0032	0.0015	0.01	0	0.083	0.059	3.8
10	7.8×10^{17}	1	1	0.0007	0.0006	0.0033	0.0051	0.057	0.057	3.2

Table 4. Characteristics of impacts of comets entering the atmosphere at a velocity of 20 km/s

D, km	M, g	D_{max}/D	E_s/E	M_{ae}/M	M_{ao}/M	H_e/H
0.2	6.3×10^{12}	2.5	0.84	6.2×10^{-4}	0.043	5.1
0.3	2.1×10^{13}	1.7	0.93	4.4×10^{-3}	0.036	1.5
0.5	9.8×10^{13}	1.6	0.96	3.7×10^{-3}	0.02	2.4
1	7.8×10^{14}	1.4	0.985	2.0×10^{-3}	0.0096	2.7
3	2.1×10^{16}	1	0.997	9.2×10^{-4}	2.0×10^{-3}	2.2
10	7.8×10^{17}	1	1	4.0×10^{-4}	2.7×10^{-4}	3.1

Table 5. Characteristics of impacts of asteroids entering the atmosphere at a velocity of 20 km/s

D, km	M, g	D_{max}/D	E_s/E	M_{ae}/M	M_{ao}/M	H_e/H
0.1	2.1×10^{12}	4.4	0.66			
0.2	1.7×10^{13}	2	0.95	0.0043	0.035	1.2
0.3	5.7×10^{13}	1.5	0.97	0.0053	0.026	2.2
0.5	2.6×10^{14}	1.4	0.985	0.0047	0.012	1.3
1	2.1×10^{15}	1.15	0.995	0.0023	0.0044	1.2
3	5.7×10^{16}	1	0.997	9.2×10^{-4}	8.0×10^{-4}	1.8
10	7.8×10^{17}	1	1	2.7×10^{-4}	0.001	2.7

ble, though they can be higher for impacts of large (greater than 10 km) bodies and also for oblique impacts.

Tables 5–6 give the computation results for asteroid impacts at typical velocities of 20 and 15 km/s. It can be seen that in the order of magnitude the ejection efficiency is identical to that for impacts of comets with the same velocities.

NUMERICAL SIMULATION OF IMPACTS ON MARS

The results of computations of atmospheric losses induced by impacts upon the Earth can provide us with some information for Mars, where the escape velocity is 5 km/s. Table 7 presents the atmosphere and impactor masses ejected to high altitudes with velocities of more than 5 km/s after impacts on the contemporary Earth. In the first approximation, the masses correspond to those that could be lost from Mars if it had a terrestrial atmosphere. These atmospheric losses on Mars are obviously much higher than on the Earth. However, the Martian atmosphere is characterized by a different composition, temperature, and scale height. Since even simulation of vertical impacts, especially of small bodies, is very time-consuming, it is not rational to perform computations for all possible atmospheres (with different masses, compositions, and temperatures), which could exist during evolution of terrestrial planets. But it is reasonable to take a limited number of computations as reference points and to try to construct approximate formulas. In the present study, we performed numerical simulations of impacts for hypothetical early atmospheres of Mars with masses greater by a factor of 10 and 100 than the mass of the present-day Earth's atmosphere (surface pressures of 3.8 and 38 bar) and with a scale height of 15 km, which corresponds to a temperature higher than that of the present Earth's atmosphere. The equation of state was taken in table form both for the air (Kuznetsov, 1965) and for carbon dioxide (Kalitkin et al., 1972; Suris, 1985). This allowed us to estimate how the results depend on the atmospheric

Table 6. Characteristics of impacts of asteroids entering the atmosphere at a velocity of 15 km/s

D, km	M, g	D_{max}/D	E_s/E	M_{ae}/M	M_{ao}/M	H_e/H
0.2	1.7×10^{13}	2	0.95	4.1×10^{-5}	4.9×10^{-3}	3.9
0.3	5.7×10^{13}	1.5	0.97	2.1×10^{-4}	7.5×10^{-3}	3
0.5	2.6×10^{14}	1.4	0.985	3.9×10^{-4}	6.8×10^{-3}	2.3
1	2.1×10^{15}	1.15	0.995	4.4×10^{-4}	2.9×10^{-3}	2
3	5.7×10^{16}	1	0.999	3.8×10^{-4}	7.7×10^{-4}	1.8
10	7.8×10^{17}	1	1	1.4×10^{-4}	3.1×10^{-4}	2.4

Table 7. The atmospheric mass (M_{am}) and the impactor mass (M_{pm}) ejected to high altitudes with velocities of more than 5 km/s after the impact as a function of the body velocity and diameter

	Comets						Asteroids			
	V = 50 km/s		V = 30 km/s		V = 20 km/s		V = 20 km/s		V = 15 km/s	
D, km	M_{am}/M	M_{pm}/M	M_{am}/M	M_{pm}/M	M_{am}/M	M_{pm}/M	M_{am}/M	M_{pm}/M	M_{am}/M	M_{pm}/M
0.075	8.4	0.1								
0.1	7.65	0.18								
0.15	2.1	0.33	0.46	0.075						
0.2	1.15	0.49	0.57	0.30	0.2	0.12	0.13	0.07	0.085	0.0022
0.3	0.7	0.77	0.32	0.33	0.15	0.13	0.11	0.059	0.053	0.029
0.5	0.5	0.87	0.16	0.41	0.098	0.12	0.046	0.055	0.023	0.015
1	0.29	0.78	0.057	0.52	0.025	0.1	0.016	0.035	0.011	0.01
3	0.046	0.87	0.012	0.48	0.0056	0.067	0.0039	0.019	0.0028	0.0072
10	0.014	0.87	0.0065	0.47	0.002	0.031	0.004	0.013	0.0028	0.0029

composition. The impact velocities were 7, 15, and 30 km/s.

The regions occupied by the atmospheric gas ejected during the impact of a 10-km body are shown in Fig. 2. If the atmospheric surface pressure is $P = 0.38$ bar, the ejection region is approximately a frustum cone, which is typical of impacts of large bodies. If the atmosphere is more massive, the impact of a 10-km body demonstrates features typical for falls of smaller bodies upon the present-day Earth, and the ejection region becomes close to cylindrical. At $P = 38$ bar, the atmospheric ejection is most efficient for impacts of bodies with a size of ~1 m. The results obtained for carbon dioxide and air atmospheres with similar pressures and scale heights proved to be rather close.

APPROXIMATION FORMULAS

An approximate formula for the atmospheric losses is given in (Svetsov, 2000). It was obtained under the assumption that a meteoroid, whose effective diameter grows with decreasing distance to the planetary surface due to fragmentation and pressure at the face, heats the atmospheric column, which later expands upward as a plane layer of a uniformly heated gas to vacuum. This is a rather crude model, but it reflects the basic mechanism of air losses arising during its upward acceleration. That is why we saved the structure of the corresponding formula when approximating the results obtained in this study and introduced additional coefficients, which were chosen so that the best accuracy was achieved for bodies smaller than 1 km. Here, we present approximation formulas containing several coefficients that can be useful for the further development of the theory of atmospheric evolution and construction of corresponding models (they can be skipped if one is not interested in the mathematical aspects of the problem).

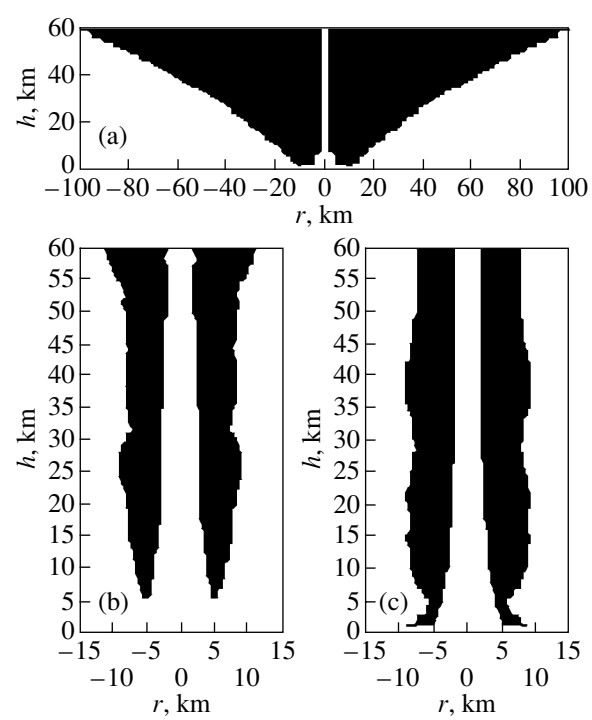

Fig. 2. The pre-impact position of the lost mass of the atmospheric gas (ejection region) for the fall of a 10-km asteroid with a velocity of 15 km/s on Mars with the atmospheric pressure equal to 0.38 bar (a), 3.8 bar (b), and 38 bar (c).

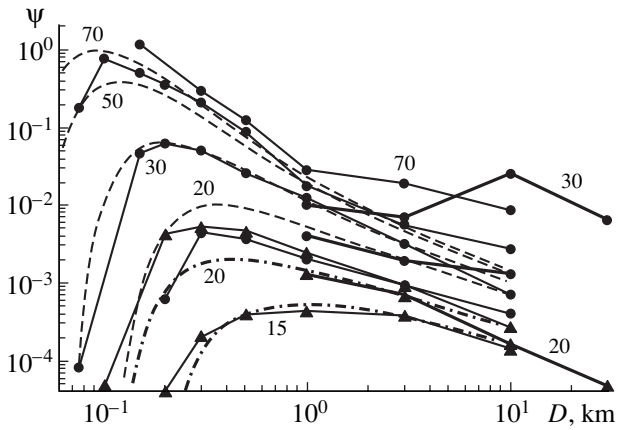

Fig. 3. The ratio of the lost atmospheric mass to the mass of the impactor as a function of its diameter and entry velocity, which is shown in km/s near the curves. Computational results are shown by solid lines. The thin lines demonstrate the results of the present study and the thick lines are for the results of (Shuvalov and Artemieva, 2000, 2002). The results obtained for comet and asteroid impacts are marked by circles and triangles, respectively. The dash-and-dot lines are approximations for asteroid impacts, while the dashed lines are approximations for comet impacts.

The ratio of the lost atmospheric mass to the impactor mass $\psi_1 = M_{ae}/M$ is approximated by the following formula:

$$\psi_1(r_0) = C_1\frac{\rho_0}{\rho_m}\left(\frac{H}{r_0} + \frac{4H^2\rho_0^{0.5}}{3r_0^2\rho_m^{0.5}} + \frac{2H^3\rho_0}{r_0^3\rho_m}\right) \times f\left(C_2\frac{V_e}{V}\exp\left(C_3\left(\frac{He_0^{0.5}}{2r_0V}\right)^{C_4}\left(\frac{V}{V_e}\right)^{C_5}\right)\right), \quad (1)$$

where ρ_0 is the atmospheric density at the planetary surface, ρ_m is the meteoroid density, H is the scale height, r_0 is the initial radius of the meteoroid, V is the entry velocity, V_e is the escape velocity, e_0 is the specific internal energy of the atmospheric gas, and the function f has the form

$$f(x) = \int_x^1 (1-\xi^2)^5 d\xi\left(\int_0^1 1-\xi^{25}d\xi\right)^{-1}. \quad (2)$$

The coefficients best fitting the computations are

$$C_1 = \frac{V_{eE}}{V_e}, \quad C_2 = 0.27\delta^{-0.21}, \quad C_3 = 1.32\delta^{0.12}, \quad C_4 = 0.27\delta^{-0.24}, \quad C_5 = 0.45\delta^{0.087}, \quad (3)$$

where $V_{eE} = 11.2$ km/s, $\delta = M_a/M_{aE}$, M_a is the atmospheric mass, and M_{aE} is the mass of the present-day Earth's atmosphere. In contrast with the corresponding formula from the study (Svetsov, 2000), an additional exponential multiplier is introduced to the argument of function f. It limits the value ψ_1 for small bodies, when the wake becomes narrow and prevents from the ejection of the heated gas to the upper atmospheric layers.

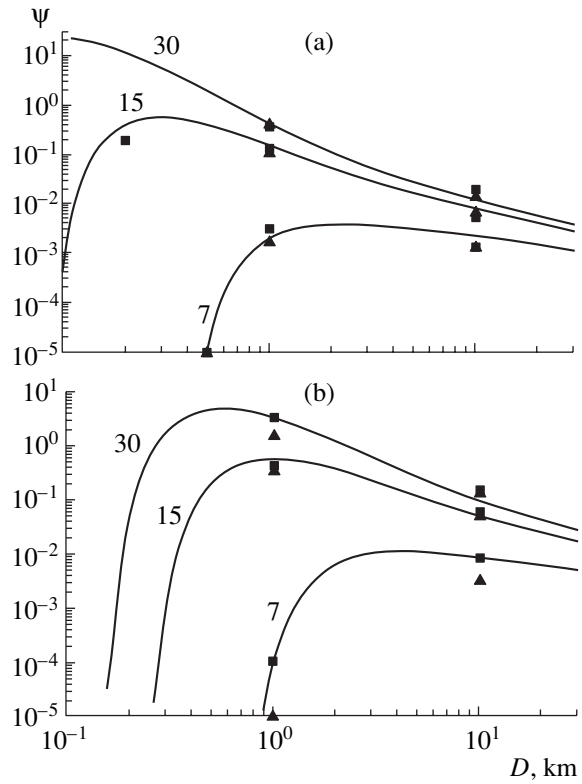

Fig. 4. The solid lines show the approximation of the ratio of the lost atmospheric mass to the mass of the impactor as a function of its diameter for the Martian atmosphere, which is by a factor of 10 (a) and 100 (b) more massive than the present atmosphere of the Earth. The entry velocities are given near the curves in km/s. The squares correspond to the numerical results for the atmosphere composed of air, and triangles are for the carbon dioxide atmosphere.

Figure 3 shows how Eq. (1) fits the computational results for the contemporary Earth's atmosphere. Note that a slight difference between the results for comets and asteroids at $V = 20$ km/s can follow from the fact that different types of equations of state were used for the substances of comets and asteroids. The computations of Shuvalov and Artemieva give a rather noticeable difference in the results obtained by them for comets and asteroids, though different equations of state (ANEOS for asteroids and equations in Tillotson's form for comets) were also used in this case. Our approximation formulas are closer to the results of Shuvalov and Artemieva. Apparently, simulations performed with a lack of information about the properties of the impactor substance, state, and shape give results differing by a factor of 2.

The approximation formula is compared to the computations for hypothetical Martian atmospheres at $\delta = 10$ and $\delta = 100$ in Fig. 4. The coefficients in Eq. (1) were chosen so that they provide the best fit both for

this case and for the modern Earth's atmosphere for an escape velocity of 5 km/s.

To construct the erosion–replenishment balance of the atmosphere, it is also necessary to know how much substance is lost by the impactor due to its acceleration to velocities above the escape one. If the impactor is sufficiently large, we can neglect the atmosphere, and the fraction of the substance lost from the impactor in the case of a homogeneous target will be the same for all bodies with the same composition. The results of computations for the impact upon an atmosphereless planet can be approximated by the formula

$$\zeta_v(V) = C_6\left(\frac{V}{V_e} - C_7\right)^{C_8}, \tag{4}$$

where ζ_v is the ratio of the lost mass of the impactor to its initial mass; $C_6 = 0.13$, $C_7 = 1.3$, and $C_8 = 1.55$ for comets and $C_6 = 0.05$, $C_7 = 2$, and $C_8 = 1.4$ for asteroids. The quantity ζ_v is bounded from above by unity, and it is assumed to be zero at $V/V_e < C_7$. It is worth noting that approximation (4) was constructed with the aim of approximating the transition region from $\zeta \ll 1$ to $\zeta = 1$ in the best way. The accuracy is lost in the region $\zeta < 0.1$. Thus, according to the computations (Svetsov, 2005), about 1% of the impactor mass is ejected during impacts of very large asteroids with a velocity of 15 km/s, but there are no losses according to Eq. (4). However, this is of no importance in our case, because almost the entire mass of the impactor is left on the planet. Note that Eq. (4) should not be used for problems in which the region $\zeta < 0.1$ is important, for example, when the impact velocities are relatively low and the mass ejected to orbits is taken into account (Vityazev and Pechernikova, 1996).

The results of computations of the lost impactor mass can be approximated as follows:

$$\zeta(r_0) = \psi_1(0.35r_0)\delta^{-1}\left(\frac{C_7V_e}{V}\right)^{0.25} + \zeta_v\frac{2}{\pi}\arctan\left(\zeta_v^4\left(\frac{r_0}{H}\right)^{1.2}\left(\frac{\rho_m}{\rho_0}\right)^{0.5}\right). \tag{5}$$

Here, the first term reflects the fact that, if the impactor is rather small, its heated substance is ejected along the wake like heated air. The second term ensures the limit value of [zeta] for large impactors. The value [zeta] is bounded from above by unity. The approximation is rather formal, since the sector approximation does not work and we still failed to construct a physical analytic model which could approximate the results with an acceptable accuracy.

Formula (1) was constructed to approximate impacts of relatively small bodies. If a body is large and its velocity is sufficiently high, the vapor after the impact propagates not only upward along the wake, but also sideways, and ejects the atmospheric gas which occupied the frustum cone before the impact. It can be concluded from the results of (Shuvalov and Artemieva, 2002) that the radius of the cone base at the Earth's surface is approximately 2.5 r_0 and the tangent of its half-angle is about 1.5. The ratio of the lost atmospheric mass to the impactor mass in this case is approximated by the formula

$$\psi_2 = \frac{3\rho_0(6.25r_0^2H + 7.5r_0H^2 + 4.5H^3)}{2\pi\rho_m r_0^3} \times \arctan\left((\zeta_v(V_i) + 10^{-3}f(0.9V_e/V_i))\left(\frac{r_0}{H}\right)^4\left(\frac{\rho_m}{\rho_0}\right)^{0.5}\right). \tag{6}$$

Here, the first term is the ratio of the air mass in the cone to the impactor mass. The velocity of the impact upon the surface V_i is determined by the formula (Svetsov, 2000)

$$V_i = V\exp\left(-\frac{\rho_0}{\rho_m}\left(\frac{H}{2r_0} + \frac{2H^2}{3r_0^2}\left(\frac{\rho_0}{\rho_m}\right)^{0.5} + \frac{H^3\rho_0}{r_0^3\rho_m}\right)\right). \tag{7}$$

For a vertical impact, the losses of the atmospheric mass can be presented as the sum $\psi = \psi_1 + \psi_2$.

Modeling of impacts oblique to the Earth's surface requires computer time an order of magnitude greater than in the case of axisymmetric vertical impacts, both because the dimension of the problem increases and because the atmospheric region covered by the flow expands. Here we present the simplest model for oblique impacts. We assume that the impacts at the angle α with respect to the vertical can be described as vertical impacts but with the atmospheric scale height increased inversely proportional to $\cos\alpha$. The computations of (Shuvalov and Artemieva, 2002; Trubetskaya and Shuvalov, 2003) for oblique impacts of 10- and 100-km asteroids show that, at an impact angle of 45°, the lost atmospheric mass increases by an order of magnitude as compared to the vertical impact. Therefore, we multiply ψ by the factor $(1 + 2\sin 2\alpha)^2$.

Let us determine the average values of the relative lost atmospheric mass η and the fractions of the remaining masses of impactors χ as follows:

$$\dot{M}_k\eta_k = \iiint \psi m\dot{n}_k(m)f_k(v)\sin 2\alpha\, d\alpha\, dv\, dm, \tag{8}$$

$$\dot{M}_k\chi_k = \iiint (1-\zeta)m\dot{n}_k(m)f_k(v)\sin 2\alpha\, d\alpha\, dv\, dm, \tag{9}$$

where $\dot{M}_k$ is the mass flux of the bodies falling onto the planet, the subscript k indicates the impactor type (asteroids, comets), $\dot{n}_k$ is the mass spectrum, f_k is the distribution of impactors in velocities v, and $\sin 2\alpha$ is the function of the angle distribution (see, e.g., Svetsov, 2002). The integrals are taken over the angle α from 0 to $\pi/2$, over the velocity v within the range of its variations, and over the mass m from 0 to the maximal impactor mass m_1 (which depends on the lifetime of the planet).

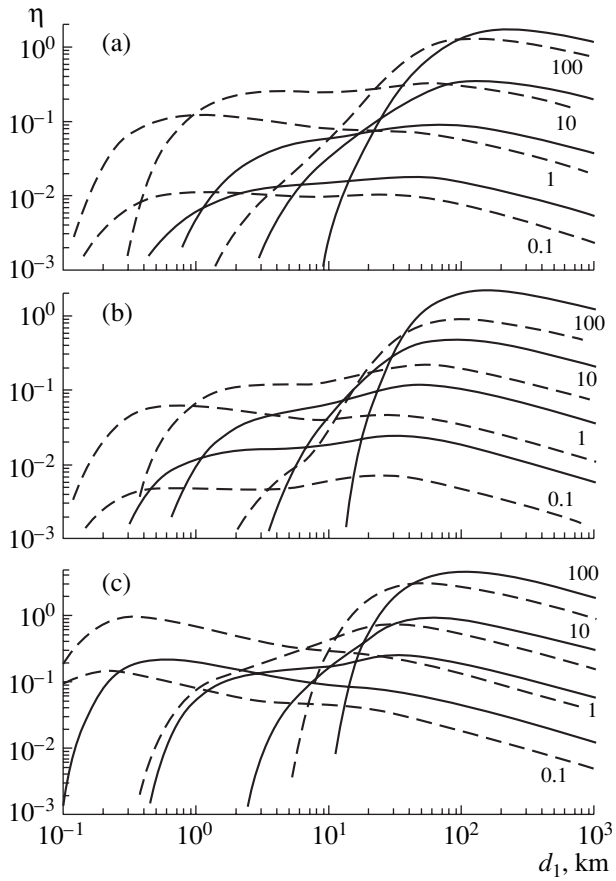

Fig. 5. The average relative lost atmospheric mass as a function of the maximal impactor size and the atmospheric pressure for impacts of asteroids (a), short-period (b), and long-period (c) comets upon the Earth (dashed lines) and Mars (solid lines). The ratio of the atmospheric mass to the mass of the present Earth's atmosphere is indicated near the corresponding curves.

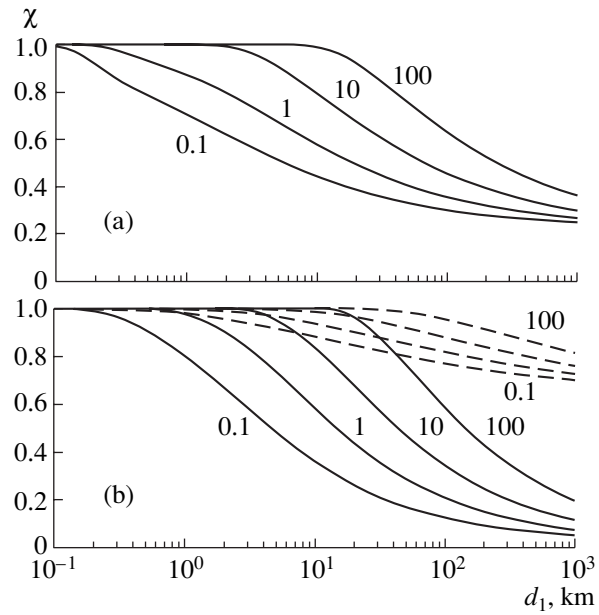

Fig. 6. The average part of the impactor mass left on the planet as a function of the maximal impactor size and the atmospheric pressure for impacts of short-period (dashed lines) and long-period (solid lines) comets upon the Earth (a) and Mars (b). The ratio of the atmospheric mass to the mass of the present Earth's atmosphere is indicated near the corresponding curves.

The well-known simple power relation (Safronov, 1969; Vityazev et al., 1990) is applied for the mass spectrum:

$$\dot{n}_k(m) = \frac{(2-q)\dot{M}_k m^{-q}}{m_1^{2-q}}, \qquad (10)$$

where $1.5 < q < 2$. The velocity distributions for asteroids ($k = 1$) and short-period ($k = 2$) and long-period ($k = 3$) comets falling onto the Earth and Mars are taken from (Jeffers et al., 2001) and (Steel, 1985; Olsson-Steel, 1987; Zahnle, 1993), respectively.

The values of η calculated by the approximate formulas as a function of the diameter of the impactor with the maximal mass m_1 and the atmospheric pressure P are shown in Fig. 5 for $q = 1.8$. If the atmospheric mass is on the same order of magnitude as the mass of the present-day Earth's atmosphere M_{aE} or less, the impacts of asteroids and short-period comets with diameters from 1 to 100 km upon the Earth on average lead to relative atmospheric losses which rather weakly depend on the impactor size. Thus, η is the value appropriate to describe the process of impact erosion of the atmosphere. The relative atmospheric losses gradually fall while the impactor size grows to 100 km or more. The value of η abruptly decreases when the impactor becomes smaller than 100 m in size. For long-period comets, whose velocities are on average much greater, such a decrease occurs for smaller impactors, and the values of η are about an order of magnitude higher than for impacts of short-period comets and two orders of magnitude higher than for asteroid impacts. The significance of large bodies increases as the atmospheric mass grows up to $100M_{aE}$. As can be seen in Fig. 5, the maximal atmospheric mass with respect to the impactor mass is ejected in this case by bodies about 100 km in size, and very small amounts of the atmospheric gas are ejected by 10-km bodies. Similar regularities are found for Mars, but the mass of the atmospheric gas ejected due to impacts of large bodies is greater than on the Earth at the same atmospheric pressures because of the smaller gravity. Conversely, smaller masses are ejected by impacts of small bodies as the scale height and the density of the atmosphere are greater on Mars.

The functions χ are demonstrated in Fig. 6. Almost the entire asteroid mass is left on Mars or the Earth after impacts on them. A large part of the mass of short-period comets is also left on the Earth and Mars (almost the entire mass on the Earth). Only after impacts of large long-period comets is a considerable portion of

their mass ejected due to their high velocities and relatively low densities. The dependences $\chi(d_1)$ are rather simple: $\chi = 1$ for small impactors and the values of χ decrease with increasing d_1 to some limiting value, which is determined by the losses of the impactor material on the atmosphereless planet and therefore is nearly independent of size.

IMPACT-INDUCED EVOLUTION OF ATMOSPHERES OF TERRESTRIAL PLANETS

Zahnle (1993) constructed a model of the evolution of the early Martian atmosphere with allowance for the influx of atmophile elements delivered by asteroids and comets and their removal due to impacts. The amount of the lost gas was determined by the tangent-plane model. Let us consider a similar model of the evolution of the early atmospheres of the Earth and Mars using the approximations obtained in our study to calculate the atmospheric losses. Neglecting degassing and absorption of the atmospheric gas by the planetary surface and assuming that all volatiles escape from the body during the impact, we can write down the following equation for mass variations of the one-component atmosphere:

$$\frac{dM_a}{dt} = \dot{M}\sum_k(\chi_k y_k - \eta_k)\beta_k, \qquad (11)$$

where t is the time, y_k is the content of atmophile elements in the impactors, and $\beta_k = \dot{M}_k/\dot{M}$ is the fraction of impactors of type k in the total mass flux $\dot{M}$.

The maximal impactor mass $m_1(t)$ can be approximately related to δM, the mass accumulated by the planet from the time t to the moment T when it stops growing (Wetherill, 1975; Zahnle, 1993)

$$m_1 = \left(\frac{4-2q}{3-q}\right)\delta M = \left(\frac{4-2q}{3-q}\right)\int_t^T \dot{M}(t)\mathrm{d}t. \qquad (12)$$

Then, solving Eq. (11) numerically, we can find M_a as a function of δM or, which is more convenient, as a function of the thickness of the accumulated substance layer $\delta h = \delta M/(4\pi R_p^2 \rho_p)$, where R_p is the planet's radius and $\rho_p = 3$ g/cm^3. Note that the values δM and δh decrease with increasing t.

Let us consider the simplest case, when the flux is composed of stony bodies only. For most variants, we take the stationary solution of Eq. (11) with a zero left-hand side at a certain value $M_a = 100$ km as the initial value of M_a. Figures 7a and 7b demonstrate pressure variations at the Earth's surface $P(\delta h)$ obtained under different assumptions. The value $q = 1.8$ corresponds to the theoretically based distribution, which is set when the bodies collide and disintegrate into fragments. However, other values of the exponent are possible (Vityazev et al., 1990). The value $y_1 = 0.01$ follows from the assumption that carbonaceous chondrites make 20% of stony bodies and the main atmophile element is carbon dioxide, which all carbon from the impactor goes into (Zahnle, 1993). (It is assumed in this case that oxygen is rapidly and completely absorbed by condensed particles, which form regolith (Mukhin and Gerasimov, 1989); water is condensed, and the concentrations of other components, including nitrogen, are considerably lower than the CO_2 concentration.) Curves *1* correspond to this basic variant ($q = 1.8$, $y_1 = 0.01$, and $\delta h_0 = 100$ km) in Figs. 7a and 7b. Since within the framework of the model suggested here the computation error can be mainly connected with inaccurate determination of the lost atmospheric mass, the time variations of atmospheric masses were computed for increased and reduced values of η (curves *2* and *3* in Fig. 7a). With η varying by a factor of 3, the final atmospheric pressures change by a factor of approximately 5. Therefore, the value of η should be refined in further studies.

A comparison of curves *1*, *4*, *5*, and *6* shows how the initial data affect the final pressure. The final atmospheric mass is mainly determined by the period $30 < \delta h < 100$ km, when the planet acquires ~99% of its mass. No matter what has happened before, the final pressure does not depend on the initial data taken at greater δh_0, and later the mass flux falls. Any substantial impact-induced changes in the mass of the early atmosphere are unlikely during the late bombardment period 3.9 to 4.0 billion years ago, when, according to the estimates of Kring and Cohen (2002), about 10^{24} g ($\delta h \sim 0.6$ km) of asteroid material fell down onto the Earth. If the atmosphere disappeared for some reason at $\delta h_0 \le 1$ km, its mass already cannot grow to the values set before. The fraction of carbonaceous chondrites in the total body flux chosen in this study corresponds to modern observations. But this fraction and the composition of carbonaceous chondrites could be different at the stage of accumulation; therefore, we also performed computations for $y_1 = 0.003$ and $y_1 = 0.001$ (curves *7* and *8*). The decrease in y_1 by an order of magnitude decreases the final pressure by a factor of more than 20. And, at last, we examined how the exponent q in the mass distribution of asteroids affects the result. The increase in q up to 1.95 (curve *10* in Fig. 7b) leads to a negligible decrease in the final pressure, but the fall of q down to 1.5 (curve *9*) gives a fourfold pressure. Thus, both the accuracy of η and the quantities q and y_1 considerably affect the solution $P(\delta h)$.

The results of similar computations for the Martian atmosphere are presented in Figs. 7c and 7d. The same regularities are found for Mars, but the final atmospheric pressure is smaller by a factor of 3 than that on the Earth. If we vary all input parameters in reasonable ranges, the final atmospheric pressure is higher than or about 1 bar. This result is completely different from those of Zahnle (1993), who considered that the entire early atmosphere of Mars was lost due to impacts.

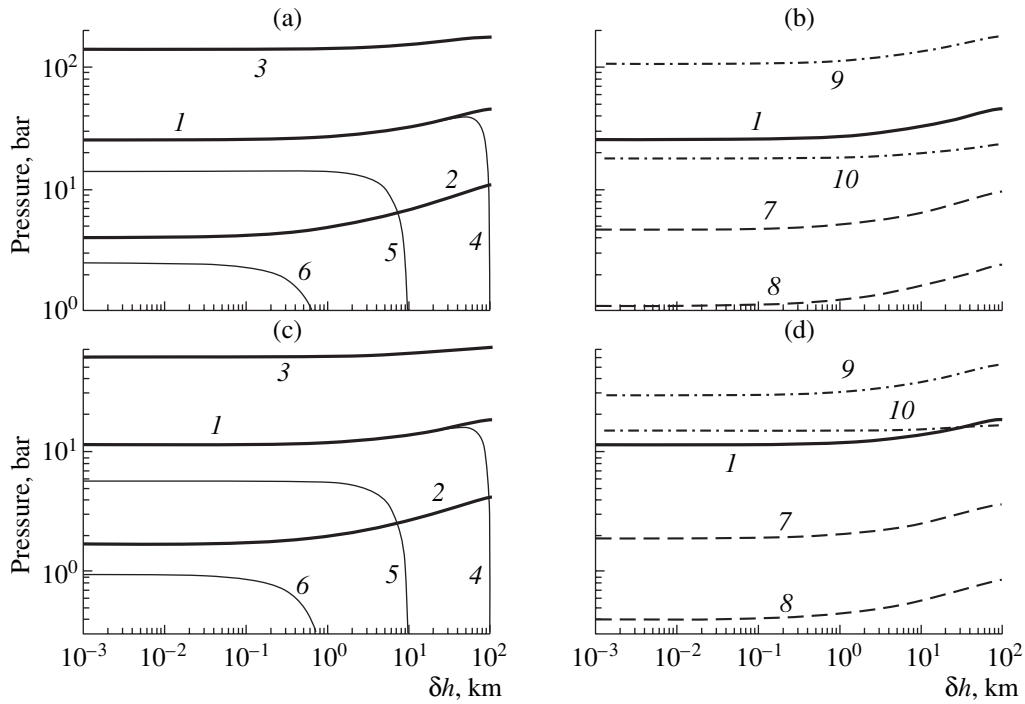

Fig. 7. Variations of the atmospheric pressure on the Earth (a, b) and Mars (c, d) in the case of the flux composed entirely of stony bodies. The solid lines are for $q = 1.8$, $y_1 = 0.01$ with different initial data: the stationary solution at $\delta h_0 = 100$ km (*1,* 2, and *3*), $P = 0$ at $\delta h_0 = 100$ km (*4*), $P = 0$ at $\delta h_0 = 10$ km (*5*), and $P = 0$ at $\delta h_0 = 1$ km (*6*). Curves *2* and *3* are respectively for η increased and decreased by a factor 3. The dashed lines are for $q = 1.8$, $y_1 = 0.003$ (*7*) and $y_1 = 0.001$ (*8*). The dash-and-dot lines are for $y_1 = 0.01$, $q = 1.5$ (*9*) and $q = 1.95$ (*10*).

The amount of cometary substance received by the Earth is unknown. Different authors give estimates varying by several orders of magnitude (see Oro and Lazcano, 1996); the maximal estimates are 10^{26} g, which is much greater than the present mass of the ocean equal to 1.4×10^{24} g. Measurements of the isotopic compositions of hydrogen in comets Halley, Hyakutake, and Hale–Bopp showed that such comets could be responsible only for 20% of oceanic water (Abe et al., 2000; Owen and Bar-Nun, 2000). But it is still unknown to which extent these comets resemble the ones that fell onto the Earth during its accumulation. Investigations of comet C/1999 S4 (LINEAR) showed that it was formed closer to the Sun than the previously studied comets and its isotopic composition of hydrogen is probably the same as the water composition on the Earth (Mumma et al., 2001).

According to the estimates of Olsson–Steel (1987), the proportion of large impact craters formed by asteroids and short-period and long-period comets is 0.82:0.12:0.06 on the Earth, 0.94:0.04:0.02 on Mars, and 0.70:0.23:0.07 on Venus. Based on these estimates, we assumed that comets made up 18% of the mass flux of impactors at the late stage of the Earth's accumulation. Then we found that 20% of the ocean mass could be delivered by the flux of comets if they started to fall at $\delta h = \delta h_c = 1$ km. Let us perform computations with the two values $\delta h_c = 1$ km and $\delta h_c = 10$ km corresponding to the moments when the cometary bombardment started, and let us adopt the ratio of the mass fluxes of different bodies within the proportions given above. The content of atmophile components in comets is not known for sure. According to (Greenberg, 1998), a typical comet must contain about 26 mass % of elements like Si, Mg, Fe, 26–30% of water, and the rest of the comet must be composed of carbon-bearing volatile molecules like CO, CO_2, CH_3OH, organic molecules, and hydrogen (~10%). We assumed the content of atmophile elements $y_2 = y_3 = 0.3$ as the most probable variant and also performed computations for $y_2 = y_3 = 0.1$ and $y_2 = y_3 = 1$.

Figure 8 demonstrates possible changes of the atmospheric pressure in the case when comets are added to the flux of stony bodies in different situations at $q = 1.8$ (except for the only variant for $q = 1.5$). Curves *1* in Figs. 8a and 8c correspond to the rather low content of the atmophile component, $y_2 = y_3 = 0.1$ (which is equivalent to a decrease in the comet flux by a factor of 3 in the variant with $y_2 = y_3 = 0.3$). In this case, comets negligibly affect the atmospheric mass if atmophile elements are delivered in sufficient quantities by chondrites ($y_1 = 0.01$). The behavior of curves *2*, *3*, *4*, and *5* in Figs. 8a and 8c shows that at $y_2 = y_3 \geq 0.3$ the major, or at least considerable, part of the atmosphere is brought by comets (as the final pressure is much higher if comets are available). If the average content of the

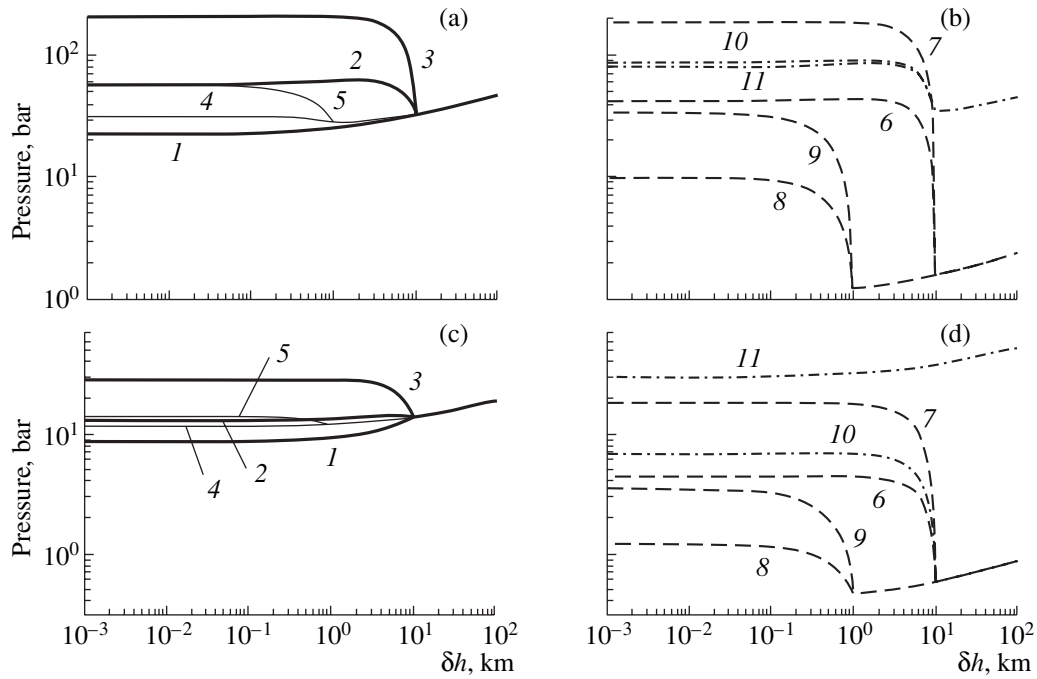

Fig. 8. Variations of the atmospheric pressure on the Earth (a, b) and Mars (c, d) in the case of the flux composed of stony bodies and comets. The solid lines are for $y_1 = 0.01$, $\delta h_c = 10$ km, $y_2 = y_3 = 0.1$ (*1*), $y_2 = y_3 = 0.3$ (*2*), and $y_2 = y_3 = 1$ (*3*); $y_1 = 0.01$, $\delta h_c = 1$ km, $y_2 = y_3 = 0.3$ (*4*) and $y_2 = y_3 = 1$ (*5*). The dashed lines are for $y_1 = 0.001$, $\delta h_c = 10$ km, $y_2 = y_3 = 0.3$ (*6*) and $y_2 = y_3 = 1$ (*7*); $y_1 = 0.001$, $\delta h_c = 1$ km, $y_2 = y_3 = 0.3$ (*8*) and $y_2 = y_3 = 1$ (*9*). The dash-and-dot lines are for the following variants. Panel (b): $y_1 = 0.01$, $\delta h_c = 10$ km, $y_2 = y_3 = 0.3$ with different mass fraction of comets in the bodies' flux: $\beta_1 = 0.18$, $\beta_2 = 0$ (*10*) and $\beta_1 = 0.23$, $\beta_2 = 0.07$ (*11*). Panel (d): $y_1 = 0.001$, $\delta h_c = 10$ km, $y_2 = y_3 = 0.3$, $\beta_1 = 0.06$, $\beta_2 = 0$ (*10*) and $q = 1.5$, $y_1 = 0.01$, $\delta h_c = 10$ km, and $y_2 = y_3 = 0.3$.

atmophile component in stony bodies is reduced to 0.001, then, as curves *6–9* in Figs. 8b and 8d show, almost the whole atmosphere is formed as a result of comet falls both at $\delta h_c = 10$ km (curves *6* and *7*) and at $\delta h_c = 1$ km (curves *8* and *9*), that is, even if they fall only at the latest stage of the accumulation. Variant *10* in Fig. 8b is the same as variant *2* in Fig. 8a but with a different mass proportion between short-period and long-period comets ($\beta_1 = 0.18$ and $\beta_2 = 0$). This variant shows that the pressure is higher with no long-period comets present; that is, long-period comets mainly promote atmospheric erosion. The same dependence is also true for Mars, which is confirmed by comparing curves *6* and *10* ($\beta_2 = 0$) in Fig. 8d. The atmospheric pressure is about 80 bar in variant *11* (Fig. 8b) with $\beta_1 = 0.23$ and $\beta_2 = 0.07$, which approximately corresponds to Venus. This pressure is a factor of 1.5 higher than that obtained in variant *2* (Fig. 8a) with $\beta_1 = 0.12$ and $\beta_2 = 0.6$. The result is nearly the same if we use gravity, the escape velocity, and the scale height of the atmosphere equal to those on contemporary Venus. Finally, comparing curve *11* in Fig. 8d for $q = 1.5$ and curve *2* in Fig. 8c for $q = 1.8$, we see that the mass distribution affects (by several times) the final atmospheric pressure. Therefore, the final atmospheric mass substantially depends on the input parameters also in the presence of the comet flux. The atmosphere turns out to be more massive when comets fall.

The value $P = 80$ bar (variant *11* in Fig. 8b) is rather close to the atmospheric pressure on Venus, where the gaseous envelope was possibly weakly eroded after the accumulation. If everything were perfect in our model, including $q = 1.8$, we could conclude that the early atmosphere (by the end of the planetary accumulation, when its mass was more than 99% of the modern one) had a pressure of about 60 bar on the Earth and of about 15 bar on Mars. However, the result depends both on q and on other input parameters and assumptions. To make the results of the model considered here more accurate, the atmospheric masses lost at high pressures due to impacts of bodies about hundreds of kilometers in size or larger should be obtained with sufficient precision. This is technically possible.

We considered the model of the early atmosphere within the framework of a one-component model under the assumption that carbon dioxide is its main component. But the process of atmospheric evolution, including the evolution of its composition, may be much more complicated. It is formally easy to construct a similar model for a multi-component atmosphere. But more reliable information about input parameters (including the content of the basic atmophile components in planetesimals and the proportion between fluxes of aster-

oids and comets of different composition) is necessary to obtain good results in practice. The problems of the chemical composition of gases escaping during impacts, their interaction with the atmosphere and rocks of the upper planetary layer, physical and chemical states of evolving early atmospheres (temperature and water vapor concentration), and the gas sink from the atmosphere to the planetary surface can be referred to as the most difficult problems of the evolution of early atmospheres. These problems were beyond the scope of the model considered here.

CONCLUSIONS

We computed the masses of the atmosphere and the impactor material lost due to vertical impacts of bodies from 100 m to 10 km in size. Approximate formulas (1)–(7) for the lost masses of the atmosphere and the impactor material were obtained based on these computations as well as on those of other authors. Although the computations do not cover the entire necessary interval of parameters of atmospheres and impacts, we made an attempt to develop a model of the evolution of early planetary atmospheres under the influence of falling bodies. It was demonstrated that bodies several hundred meters in size, which strongly disintegrate during their fall in the atmosphere, are more effective in terms of atmospheric erosion, leading to losses from the lower atmospheric layers. Nevertheless, according to the computations by the integral model of atmospheric evolution, the final parameters of the atmosphere considerably depend on the period of accumulation when the deficit of the layer of the accumulated substance varied from ~100 to ~10 km and impacts of very large bodies occurred. In this connection, to make the results more accurate within the framework of the given model, it is reasonable first to compute masses lost due to impacts of bodies larger than 100 km upon planets with massive atmospheres and estimate the state of these atmospheres.

ACKNOWLEDGMENTS

The author is grateful to N.A. Artemieva, A.V. Vityazev, I.V. Nemchinov, and V.V. Shuvalov for useful discussions.

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